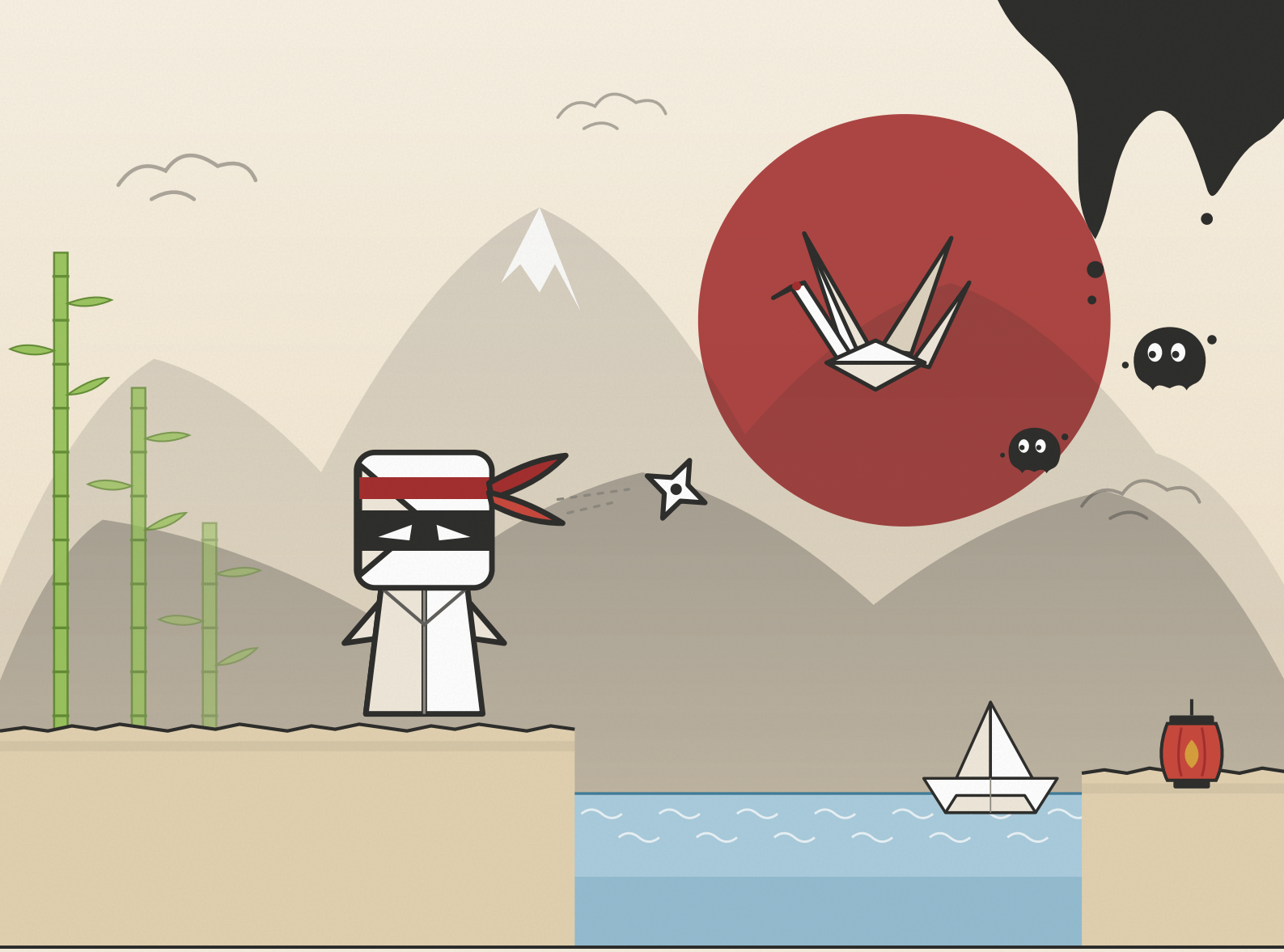


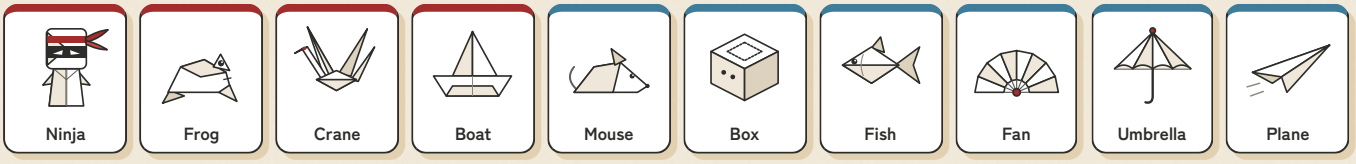
# KAMI 紙

## GAME DESIGN DOCUMENT

A paper ninja folds into animal shapes to cross a painted world, but every fold leaves a crease, and too many tear him apart.

*Fold wisely.*

10 FOLDS TO MASTER





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COURSE
Game Development with Unity · USAL

DATE
1 November 2026

GENRE
2D puzzle-platformer · PC

SECTION 1

# Overview

**Kami** is a 2D paper-ninja platformer where you fold into paper shapes (you start with a ninja, frog, crane and boat and unlock six more, like a fish, a fan and a paper plane), leap over tall walls, glide on the wind, float across rivers, and pop Ink Blots with paper stars to seal the inkwell and save a painted world. What makes it special is that every fold leaves a crease and each level allows only so many before you tear, every obstacle is a "which shape gets me through?" puzzle, pressing stones make you plan your folds ahead, and the whole world is made of paper and ink. It's for kids 8 and up and their families, casual platformer fans, and puzzle lovers who enjoy short levels.

## High concept

A paper ninja folds into animal shapes to cross a painted world, but every fold leaves a crease, and too many tear him apart.

## Genre and platform

2D puzzle-platformer · PC (Windows), keyboard · side view

## Audience

Kids 8+ and families · casual platformer fans · puzzle lovers who like 2–4 minute levels

## Unique selling points

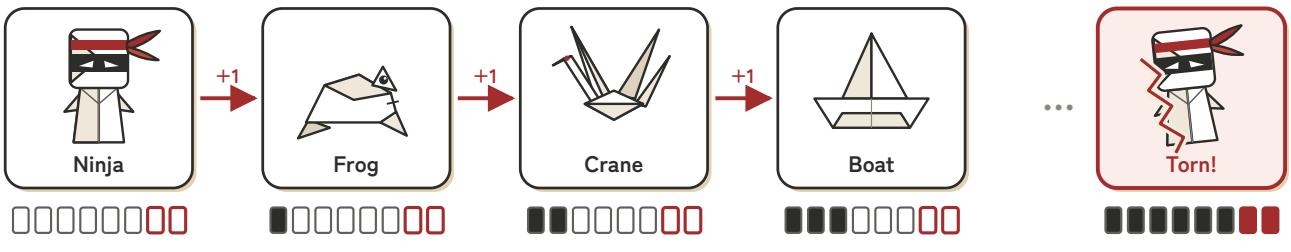
- 10 shapes to unlock, 4 on your fold belt at a time
- Creases make every fold a choice
- A world made of paper and ink
- Every obstacle is a shape puzzle

## Self-scored scorecard

| Criterion | Score | Weight | Points | Why |
|---|---|---|---|---|
| **Fun core** | ●●●●● | ×2 | **10** | A "which shape?" choice every few seconds |
| **Clarity** | ●●●●● | ×1 | **5** | "Fold into shapes, but don't tear" |
| **Hook** | ●●●●● | ×1 | **5** | The hero's weakness, creases, is the main rule |
| **Audience** | ●●●●● | ×1 | **5** | Kids 8+, families, casual platformer fans |
| **Feasibility** | ●●●●● | ×2 | **10** | 2D physics and one shape state machine |
| **Content load** | ●●●●○ | ×1 | **4** | 13 levels and 10 shape sprites |
| **Testability** | ●●●●● | ×1 | **5** | Folds, restarts and time per level |
| **Total** | | | **44 / 45** | |

**44/45**
Go range
35–45

## Kami in one picture




Every fold costs one crease. At the level's limit (8 in this example) the paper tears and Kami restarts at the last pressing stone.

Fold wisely.

SECTION 2

# Pillars and experience

Kami is built on three pillars, and every feature must serve at least one of them or it is cut.




### Every fold is a choice

Folding always costs a crease, so the player thinks before changing shape.



### Clever, not strong

Kami wins by picking the right shape, not by fighting; there is no health bar or combat upgrade.



### A world of paper and ink

Every rule comes from paper: water soaks it, fire burns it, wind lifts it, ink stains it.

## Feature test

| Feature idea | Pillar it serves | Keep or cut |
|---|---|---|
| Crease limit set per level | Every fold is a choice | Keep |
| Fold belt of 4 shapes | Every fold is a choice | Keep |
| New shapes unlocked by levels | Clever, not strong | Keep |
| Water soaks, fire burns, wind lifts | A world of paper and ink | Keep |
| Sword combos and a health bar | None | Cut |
| Long story cutscenes | None | Cut |

## MDA: what we build and what players feel

### Mechanics
what we build

- Shapes on a 4-slot fold belt
- Crease limit per level, pressing stones
- Shapes unlocked, stars for folds under par

→

### Dynamics
what happens in play

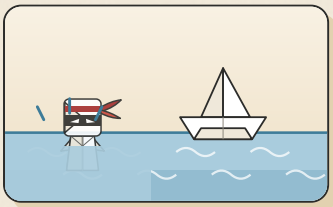
- Reading a room, then picking the shape that beats it
- Saving creases for the hard part ahead
- Replaying levels with fewer folds

→

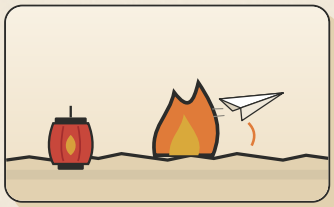
### Aesthetics
what the player feels

- Cleverness: the "aha" when it clicks
- Tension near the limit, relief at a stone
- Curiosity and pride

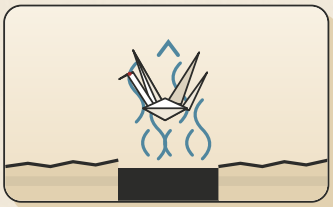
## A world of paper and ink: the rules at a glance



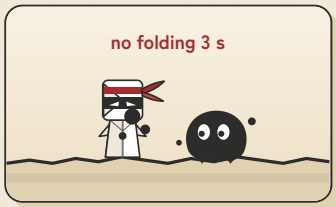
**Water soaks**
Only the boat and the fish stay afloat.



**Fire burns**
Any shape tears on touch.



**Wind lifts**
Only the crane rides it across gaps.



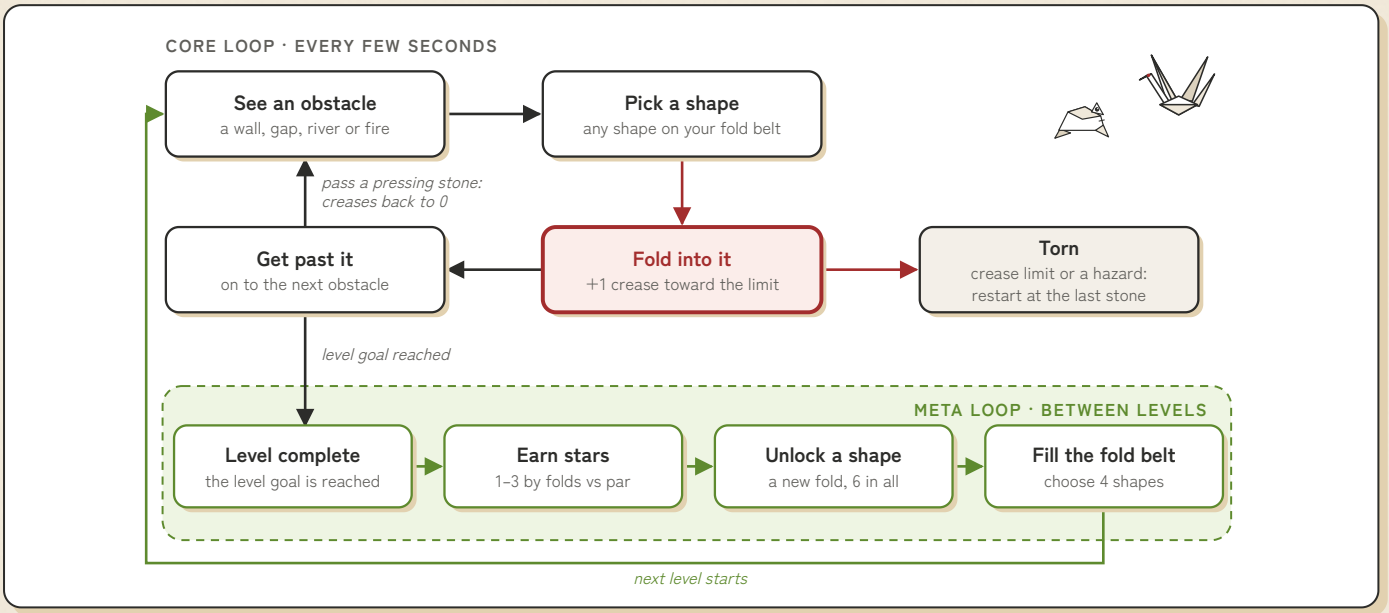

**Ink stains**
A blot's touch: no folding for 3 s.

SECTION 3

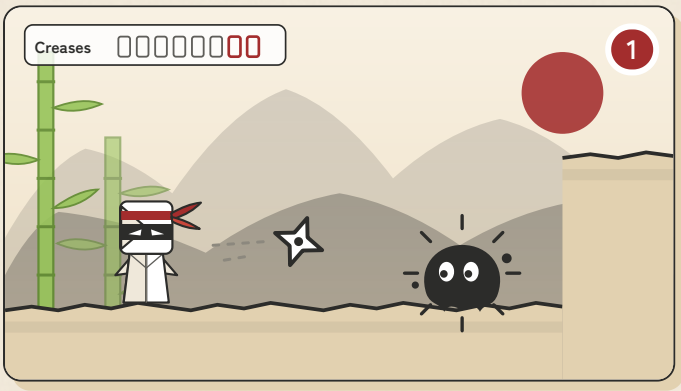
# Gameplay

In every room the player reads an obstacle, picks a shape, pays one crease to fold into it, and moves on; between levels they replay to fold less and earn stars.

## Core and meta loop

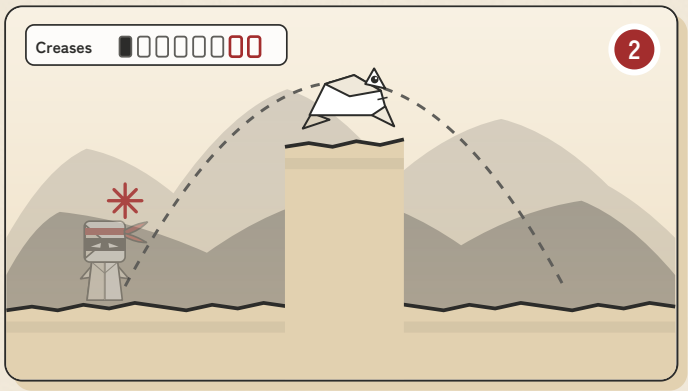



## Example: level 2, Water and wind



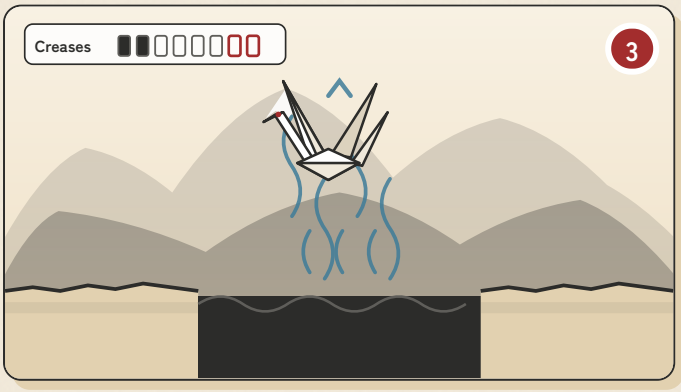

### Pop the blot (0 creases)

The ninja flicks a paper star. Ink Blots stain Kami if they touch him.



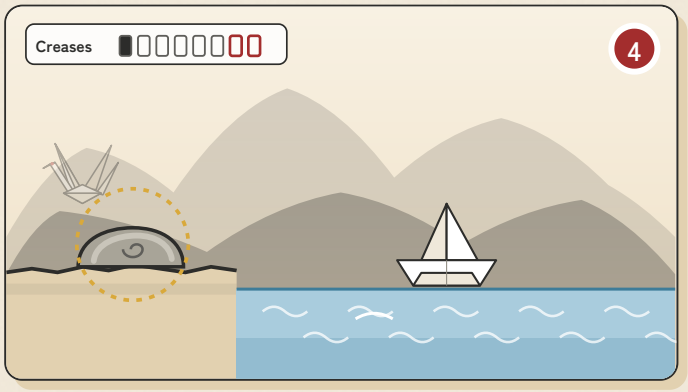

### Fold into the frog (+1 crease)

The wall is too tall for the ninja, so Kami folds into the frog and leaps it.




### Glide on the wind (+1 crease)

The gap is an ink pit. The crane rides the wind column across.




### Press, then float (reset, +1)

The pressing stone resets creases to 0, then the boat crosses the river.

SECTION 3 · GAMEPLAY, CONTINUED

## Rules (sample values, 1 unit = 1 tile)

| Rule | Value |
|---|---|
| **Crease limit** | 5–8 per level; each fold = 1 crease; reaching the limit tears Kami, who restarts at the last stone |
| **Fold belt** | 4 shapes per level on keys 1–4; the ninja always sits in slot 1 |
| **Pressing stone** | Creases back to 0, checkpoint, belt swap; 0–2 per level |
| **Paper star** | 14 tiles/s, range 8 tiles, 0.35 s cooldown, 1 hit pops a blot |
| **Ink Blot** | Patrols at 1.3 tiles/s; a touch blocks folding for 3 s |
| **Water, wind, fire** | Water sinks all but the boat and fish · wind lifts the crane at 8 tiles/s · fire tears on touch |
| **Stars** | 3 at or under par folds, 2 at par + 2, 1 for finishing |

## Shapes (sample values · speed in tiles/s, jump in tiles)



### Ninja

Starting shape

| SPEED | JUMP |
|---|---|
| 6 | 2 |

+ Paper stars
− Can't clear tall walls



### Frog

Starting shape

| SPEED | JUMP |
|---|---|
| 3.5 | 5 |

+ Leaps 5-tile walls
− Slow



### Crane

Starting shape

| SPEED | JUMP |
|---|---|
| 5 | Glides (falls 1.5/s) |

+ Rides wind
− Can't jump

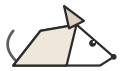


### Boat

Starting shape

| SPEED | JUMP |
|---|---|
| 1.5 land, 5 water | 0 |

+ Floats
− Almost stuck on land



### Mouse

Unlocks after L2

| SPEED | JUMP |
|---|---|
| 7 | 1 |

+ Fits 1-tile tunnels
− Tiny jump

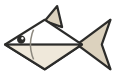


### Box

Unlocks after L4

| SPEED | JUMP |
|---|---|
| 2 | 0.5 |

+ Holds switches, blots bounce off
− Sinks



### Fish

Unlocks after L6

| SPEED | JUMP |
|---|---|
| 1 land, 6 water | 0 |

+ Swims 6 s under water
− Flops on land



### Fan

Unlocks after L7

| SPEED | JUMP |
|---|---|
| 4 | 1.5 |

+ 4-tile gust
− Can't stop big fires



### Umbrella

Unlocks after L9

| SPEED | JUMP |
|---|---|
| 4 | 1 |

+ Blocks drips, falls slowly
− Blown by wind



### Plane

Unlocks after L10

| SPEED | JUMP |
|---|---|
| 12 (dash) | 0 |

+ 10-tile dash
− Can't steer

## Controls (keyboard and mouse)

| | | | |
|---|---|---|---|
| Move | A D or ← → | Jump | W ↑ or Space |
| Fold into belt slot 1–4 | 1 2 3 4 | Shape ability (paper star, fan gust, plane dash) | F or left click |
| Swap belt shapes at a pressing stone | Tab | Restart from last stone | R |
| Pause | Esc | | |

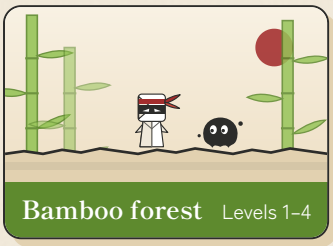
Fold wisely.

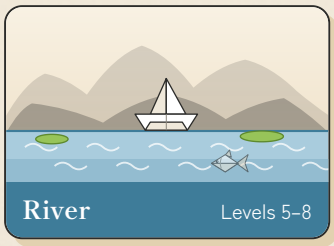
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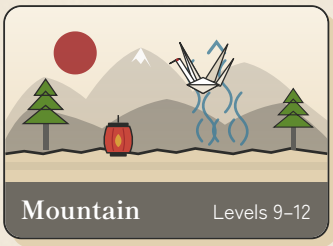
SECTION 4

# Progression and levels

Kami is level-based: 13 hand-made levels in 3 painted scrolls plus a finale. Scroll 1 teaches the 4 starting shapes, and from level 2 on, about every second level unlocks a new shape, which the next level teaches with **introduce → practice → twist → test**. Each level sets its own crease limit: 8 in the first levels, falling to 5 in the mountain test. Par folds rise from 2 to 10, so planning folds between pressing stones matters more and more.







| Level | New element | Goal |
|---|---|---|
| 1 **First steps** | Ninja and frog: paper stars, tall walls | Pop 2 blots, climb 2 walls |
| 2 **Water and wind** | Boat and crane unlocks the mouse | Cross a river and a wind gap |
| 3 **Bamboo tunnels** | Mouse through 1-tile tunnels | Find the tunnel route |
| 4 **Bamboo test** | All 5 shapes, 1 stone unlocks the box | Plan folds to the stone |
| 5 **Stone gates** | Box holds down switches | Open 3 gates |
| 6 **Fast water** | River currents unlocks the fish | Cross against the flow |
| 7 **Under the river** | Fish in underwater tunnels unlocks the fan | Swim to the far bank |
| 8 **River test** | Fan pushes lily pads and blots | Reach the bank with 1 stone |
| 9 **Mountain wind** | Long crane routes unlocks the umbrella | Ride 3 wind columns |
| 10 **Ink rain** | Umbrella under ink drips unlocks the plane | Cross the dripping cave |
| 11 **Long gaps** | Plane dashes over 10-tile gaps | Clear 3 long gaps |
| 12 **Mountain test** | Belt choice matters, 2 stones | Finish without tearing |
| 13 **The inkwell** | Rising ink chase, every shape | Reach and seal the inkwell |

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SECTION 5

# World and story

## Setting

The game takes place inside an old painted scroll: a bamboo forest, a wide river and a snowy mountain, all made of paper, brushstrokes and ink. An inkwell at the top of the scroll has tipped over, and a spill of living ink is creeping down, erasing everything it touches and spawning Ink Blots.

## Hero

Kami is a tiny ninja folded from the scroll's last blank page. He isn't strong, but he can fold himself into a frog, a crane or a boat, and he learns new folds (a mouse, a box, a fish, a fan, an umbrella and a paper plane) from lost fold diagrams hidden in the scroll. Every fold leaves a crease, and if he folds too often he tears.

## Why the player should care

If the ink reaches the bottom of the scroll, the whole painted world is gone. Kami is the only page left that can still move.



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SECTION 6

# Art and audio

Kami uses flat 2D paper-cut art (simple shapes with a paper texture), which is low to medium cost for a solo developer and makes the paper rules easy to read.

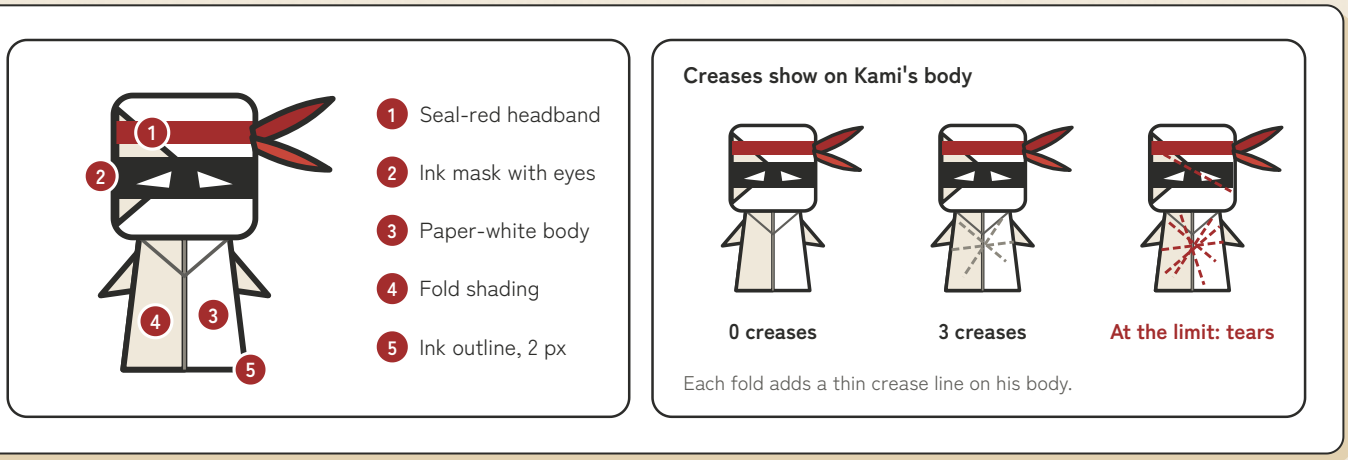
## Five adjectives

papery · hand-painted · calm · flat · warm

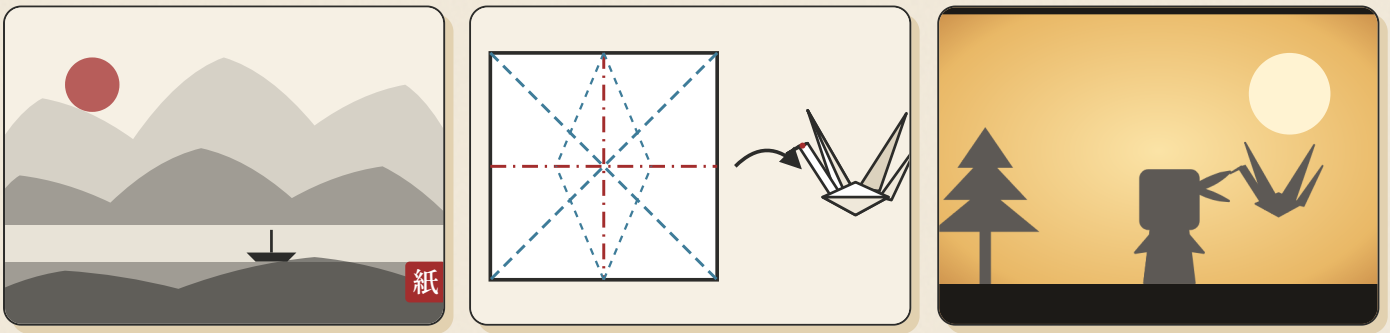
## Palette (6 colours)

| | | | | | |
|---|---|---|---|---|---|
| **Parchment** #F4ECDD Sky and background paper | **Paper white** #FFFFFF Kami and paper stars | **Ink black** #2C2C2A Ink, outlines | **Bamboo green** #97C459 Scroll 1 plants | **River blue** #A9CCDD Water and wind | **Seal red** #A32D2D Inkwell, headband |

## Kami character sheet




## 3 reference images (original mood sketches: sumi-e ink landscape · origami crane folds · paper-cut shadow puppets)




## Audio

**Music mood:** calm, wooden, curious (bamboo flute and plucked strings), with a faster drum layer for the rising-ink finale.

**SFX list** paper fold · crease warning · paper tear · hop and frog leap · wing flap and wind · splash and paddle · bubbles (fish) · soggy sink · mouse squeak · box thud and switch click · gate grind · fan gust · umbrella pop and drips · plane whoosh · paper star flick · ink pop · ink splat · fire crackle · pressing-stone chime · unlock jingle · seal stamp (level complete) · paper tap (menus)

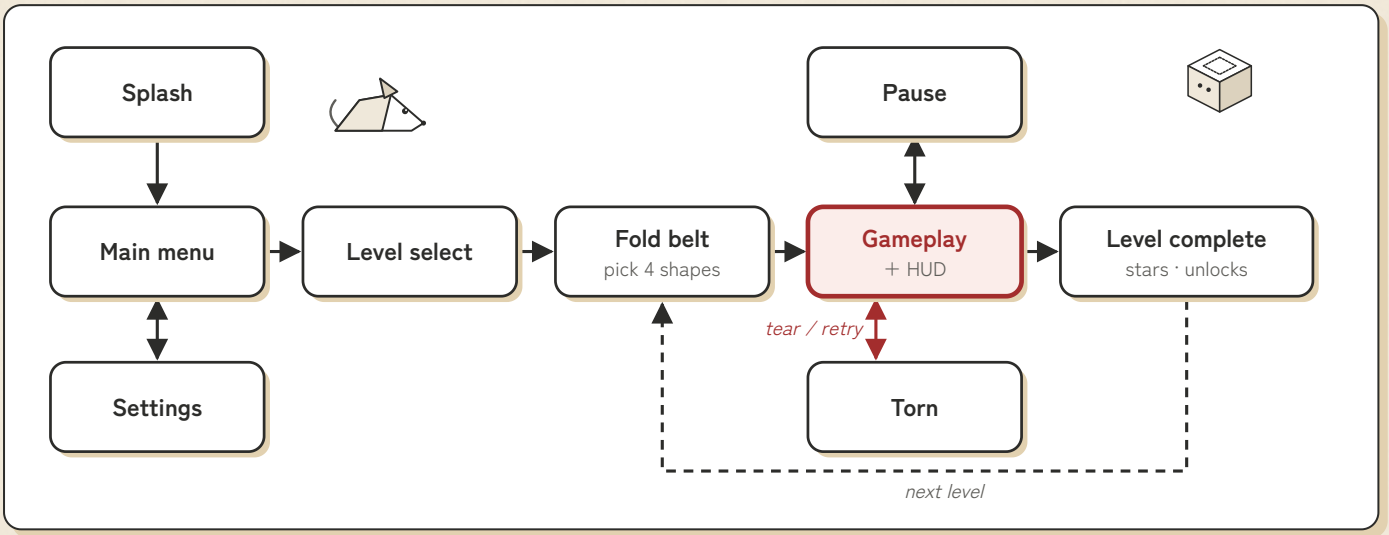
**Free assets and licences:** Kenney.nl (UI buttons, placeholder tiles; CC0) · Freesound.org (paper, water and fire sounds; CC0 or CC BY, credit the author) · Google Fonts (menu and HUD font; SIL Open Font License).

SECTION 7

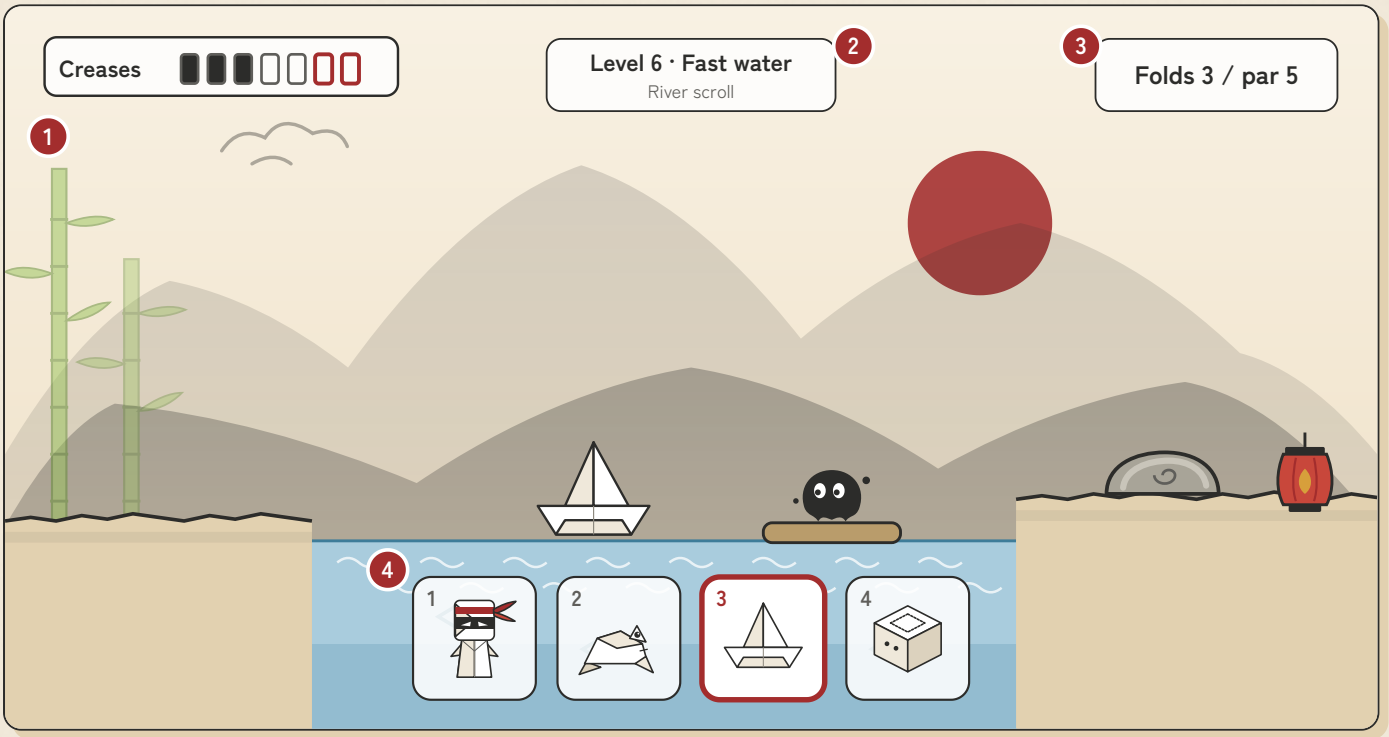
# UI and screens

Kami has 9 screens, and each one becomes one Canvas screen in Unity.

## Screen flow




## HUD (level 6 mid-play)




1 **Crease meter.** One mark per crease up to the level's limit; the last two marks are red.

2 **Level name.** Shown for 2 s at the start, then hidden.

3 **Folds and par.** Folds used so far against the level's par.

4 **Fold belt.** The 4 shapes on keys 1–4; the current shape is highlighted.

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SECTION 8

# Technical (Unity)

Kami is a 2D URP project; every tuning number lives in a ScriptableObject.

| In the GDD | In Unity | Kami example |
|---|---|---|
| **Player, enemies, items** | Prefabs | Kami, InkBlot, PaperStar, Gate |
| **Tuning numbers** | ScriptableObjects | ShapeData ×10, LevelData |
| **Mechanics and rules** | C# scripts | ShapeSwitcher, FoldBelt, CreaseCounter |
| **Levels** | Scenes, Tilemaps | Level_01 to Level_13 |
| **Controls** | Input actions | Move, Jump, FoldSlot1–4, Ability |
| **Screens** | Canvas (UGUI) | MainMenu, FoldBelt, HUD, Pause |
| **Sound** | AudioMixer groups | Music, SFX, UI |
| **Look** | 2D URP, sprite atlas | Paper-cut sprites, parallax |

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SECTION 9

# Scope and risks

### Must

7 shapes (4 starting + mouse, box, fish) · fold belt · crease limit · pressing stones · Ink Blots · water, wind, fire · levels 1–7 · menus and HUD

### Should

Fan, umbrella, plane · levels 8–13 · stars and par · full SFX

### Could

Paper patterns · gamepad · Arabic and French text

### Won't (this semester)

Multiplayer · level editor · mobile · online leaderboard

## Top three risks

| Risk | Chance | Plan |
|---|---|---|
| **13 levels and 10 shapes take too long alone** | High | Build the Must list first (7 shapes, levels 1–7) |
| **Shapes feel alike** | Medium | Give each new shape one obstacle only it beats; grey-box before art |
| **Crease limits too harsh or too easy** | Medium | Log folds and tears in playtests; tune LevelData, not code |

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SECTION 10

# Change log

Every change to the design is recorded here with its reason; entry one is the pitch-day verdict.

| Version | When | Change | Why |
|---|---|---|---|
| **v0.1** | Pitch day | Verdict: Go / Pivot, score __ / 45 | Teacher's feedback |
| **v0.2** | 3 Nov 2026 | Ninjago-based idea → original paper ninja, Kami | Avoid LEGO's characters and look |
| **v1.0** | 3 Nov 2026 | GDD v1: 4 shapes, crease limit of 10 | First full design |
| **v1.1** | 3 Nov 2026 | 10 shapes unlocked through levels, 4-slot fold belt, crease limit per level (5–8) | More variety with 4 keys; gentle early levels, tight later ones |

Fold wisely.




# Fold wisely.

Kami seals the inkwell, and the scroll's colours come back.